SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I Abhinav charles (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers.

Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of 240 MB through an IoT-enabled healthcare network. The Presentation Layer compresses the data by 30%, the Transport Layer divides the compressed data into 1400-byte segments, and the Data Link Layer adds 60 bytes of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple

Question 1

1. Calculate Total Segments and Frames

- **File Size:** $150 \text{ MB} = 150 \times 10^6 \text{ bytes} = 150,000,000 \text{ bytes}$ (or $150 \times 1,024 \times 1,024 = 157,286,400 \text{ bytes}$ if using binary prefixes. Standard networking typically uses decimal values for bandwidth calculations).
    - *Using decimal (standard network metric):* $\text{Total\ Segments}=\frac{150,000,000\text{\ bytes}}{1,500\text{\ bytes/segment}}=\mathbf{100,000}\text{\ segments}$
    - *Using binary:* $157,286,400 / 1,500 = \mathbf{104,858 \text{ segments}}$ (rounded up).
- **Total Frames:** Since each segment maps to exactly one frame, the total number of frames is equal to the number of segments: **100,000 frames** (decimal) or **104,858 frames** (binary).

2. Calculate Total Data Link Layer Overhead

- Each frame adds 40 bytes of header and trailer.
- *Using decimal segments:* $\text{Total\ Overhead}=100,000\text{\ frames}\times 40\text{\ bytes/frame}=\mathbf{4,000,000}\text{\ bytes\ (4\ MB)}$
- *Using binary segments:* $104,858\text{\ frames}\times 40\text{\ bytes/frame}=\mathbf{4,194,320}\text{\ bytes}$

3. Estimate Total Transmission Time

- **Total Data to Transmit:** Original file size + Data Link overhead.
    - *Using decimal:* $150,000,000 \text{ bytes} + 4,000,000 \text{ bytes} = 154,000,000 \text{ bytes}$.
    - Convert to bits: $154,000,000 \times 8 = 1,232,000,000 \text{ bits}$.
- **Bandwidth:** $50 \text{ Mbps} = 50,000,000 \text{ bits/second}$.
- **Transmission Time:** $\text{Time}=\frac{1,232,000,000\text{\ bits}}{50,000,000\text{\ bits/sec}}=\mathbf{24.64}\text{\ seconds}$
  (Using binary values yields $\approx 25.84 \text{ seconds}$)

4. Reliability Explanation

- **Transport Layer:** Ensures reliable, end-to-end communication using sequence numbers (to reorder packets), acknowledgments (ACKs), and retransmission timers for lost segments.
- **Data Link Layer:** Ensures reliable hop-to-hop communication over a single link by using error-detection codes (like CRC in the trailer) to identify and discard corrupted frames.

Question 2 & Question 4 (Identical)

1. Calculate Number of Transmission Windows Required

- **Total Frames:** 48 frames
- **Window Size:** 6 frames per window
- $\text{Windows\ Required}=\frac{48\text{\ frames}}{6\text{\ frames/window}}=\mathbf{8}\text{\ windows}$

2. Determine Total Number of Retransmissions

- The problem states that 1 frame in **every** window is lost.
- There are 8 initial windows, meaning 8 frames are lost during the first round of transmission.
- These 8 lost frames must be resent. They will form new windows:
  - Round 2: 8 frames require $\lceil 8 / 6 \rceil = 2$ windows. This causes 2 more losses.
  - Round 3: 2 frames require $\lceil 2 / 6 \rceil = 1$ window. This causes 1 more loss.
  - Round 4: 1 frame requires 1 window. This causes 1 final loss.
  - Round 5: 1 frame successfully transmitted (assuming a clean run or completing the chain).
- **Total Retransmissions:** $8 + 2 + 1 + 1 = \mathbf{12 \text{ retransmissions}}$.
  *(Note: If the question implies a simpler interpretation where errors only happen on the first pass of the 8 windows, the answer is **8 retransmissions**).*

3. Flow Control Explanation

- Flow control prevents a fast sender from overwhelming a slow receiver with too much data. By regulating the data flow based on the receiver's buffer capacity, it eliminates buffer overflows, reduces dropped packets, and minimizes costly retransmissions.

Question 3

1. Calculate Total Number of Fragments Created

- **Total IP Packet Size:** 12,000 bytes.
- **Total Payload Size:** $12,000 \text{ bytes} - 20 \text{ bytes (original IP header)} = 11,980 \text{ bytes}$.
- **MTU Size:** 1500 bytes.
- **Maximum Payload per Fragment:** $1,500 - 20 \text{ bytes (fragment IP header)} = 1,480 \text{ bytes}$.
- *Note: Fragment payloads must be a multiple of 8 bytes for fragmentation offset mapping. 1480 is divisible by 8 ($1480 / 8 = 185$).*
- **Number of Fragments:**
  $\text{Fragments}=\lceil \frac{11,980\text{\ bytes}}{1,480\text{\ bytes/fragment}}\rceil$

=\lceil 8.09\rceil =\mathbf{9}\text{\ fragments}\)
(8 fragments of 1480 bytes of payload + 1 final fragment of 140 bytes of payload).

2. Calculate Total Header Overhead Introduced

- Each of the 9 fragments contains a 20-byte IP header.
- \(\text{Total\ Fragment\ Headers}=9\times 20\text{\ bytes}=180\text{\ bytes}\)
- **Net Overhead Added:** Since the original packet already had a 20-byte header, the *additional* overhead introduced by fragmentation is \(180 - 20 = \mathbf{160 \text{ bytes}}\).

3. Successful Delivery Explanation

- The Network Layer uses three fields in the IP header to ensure successful assembly at the destination:
  - **Identification:** Matches all fragments belonging to the same original packet.
  - **Fragment Offset:** Tells the destination exactly where each fragment's payload fits into the original reconstructed packet.
  - **More Fragments (MF) Flag:** Set to 1 for all fragments except the last one (set to 0), letting the receiver know when all pieces have arrived.

4.. Reliability ExplanationTransport Layer: Ensures reliable, end-to-end communication using sequence numbers (to reorder packets), acknowledgments (ACKs), and retransmission timers for lost segments.Data Link Layer: Ensures reliable hop-to-hop communication over a single link by using error-detection codes (like CRC in the trailer) to identify and discard corrupted frames.

Problem 5 Solutions

1. Checkpoints Created Before Failure

- **Total data sent before failure:** \(390\text{ MB}\)
- **Checkpoint interval:** Every \(60\text{ MB}\)
- **Calculation:** \(\frac{390}{60} = 6.5\)
- **Answer: \(6\text{ checkpoints}\)** were successfully created (at \(60, 120, 180, 240, 300,\text{ and } 360\text{ MB}\)).

2. Amount of Data to Retransmit

- **Last successful checkpoint:** \(6 \times 60\text{ MB} = 360\text{ MB}\)
- **Data sent after last checkpoint:** \(390\text{ MB} - 360\text{ MB} = 30\text{ MB}\)
- **Remaining data to transmit:** \(600\text{ MB} - 360\text{ MB} = 240\text{ MB}\)
- **Answer: \(240\text{ MB}\)** must be retransmitted (the \(30\text{ MB}\) lost after the last checkpoint + the remaining \(210\text{ MB}\) of the file).

3. Importance of Synchronization in the Session Layer

- **Fault Tolerance:** Prevents restarting massive file transfers from scratch during network drops.
- **Bandwidth Savings:** Saves network resources by only sending lost data segments.
- **Time Efficiency:** Reduces overall recovery time after a connection failure.

Problem 6 Solutions

1. Compressed File Size

- **Original size:** 900 MB
- **Compression reduction:** 40%
- **Calculation:** $900 \times (1 - 0.40) = 900 \times 0.60$
- **Answer: 540 MB**

2. Encrypted File Size

- **Encryption overhead:** 5% added to compressed size
- **Calculation:** $540 \times (1 + 0.05) = 540 \times 1.05$
- **Answer: 567 MB**

3. Total Percentage Reduction

- **Total size saved:** $900\text{ MB} - 567\text{ MB} = 333\text{ MB}$
- **Calculation:** $\left(\frac{333}{900}\right) \times 100\%$
- **Answer: 37%** overall reduction.

4. Presentation Layer Performance Improvements

- **Efficiency:** Compression reduces the file size, which decreases bandwidth usage and speeds up transmission times.
- **Security:** Encryption transforms data into unreadable ciphertext, preventing unauthorized interception during transit.

Problem 7 Solutions

1. Total Number of Requests Generated

- **Total data size:** $3\text{ GB} = 3000\text{ MB}$
- **Request size:** 3 MB
- **Calculation:** $\frac{3000\text{\ MB}}{3\text{\ MB}}$
- **Answer: 1000 requests**

2. Total Transmission Time

- **Total data in bits:** $3\text{ GB} = 3 \times 1024 \times 1024 \times 1024 \times 8\text{ bits} \approx 25.77\text{ billion bits}$

- *(Note: Using decimal notation: $3\text{ GB} = 3000\text{ MB} = 3,000,000\text{ KB} = 3,000,000,000\text{ bytes} \times 8 = 24,000,000,000\text{ bits}$)*
- **Throughput rate:** $120\text{ Mbps} = 120,000,000\text{ bits per second}$
- **Calculation (using decimal base):** $\frac{24,000,000,000}{120,000,000}$
- **Answer: 200 seconds** *(or **214.75 seconds** if your curriculum utilizes binary gigabytes 2^{30})*.

3. Reliability in the Application Layer

- **Protocol rules:** Employs underlying reliable transport protocols like TCP (via FTP) to ensure error-free file delivery.
- **User interface:** Provides simple error messages and recovery options if a network transaction drops.

Problem 8 Solutions

1. Total End-to-End Delay

- **Total processing delay:** $4\text{ ms} + 3\text{ ms} + 5\text{ ms} + 2\text{ ms} + 1\text{ ms} = 15\text{ ms}$
- **Other delays:** Propagation (18 ms) + Transmission (12 ms)
- **Calculation:** $15\text{ ms} + 18\text{ ms} + 12\text{ ms}$
- **Answer: 45 ms**

2. Layer Contributions to Communication

- **Application Layer (4 ms):** Converts user network data into web, email, or file requests.
- **Transport Layer (3 ms):** Handles end-to-end connection reliability, flow control, and data segmentation.
- **Network Layer (5 ms):** Manages logical routing paths and packet addressing across networks.
- **Data Link Layer (2 ms):** Packages data into frames for physical link transmission and handles local error checks.
- **Physical Layer (1 ms):** Converts digital bitstreams into electrical, optical, or radio signals on physical media.

Problem 9 Solution

Given Data

- **Total useful data:** $80\text{ MB} = 80 \times 10^6\text{ bytes}$ (or $80 \times 1024 \times 1024 = 83,886,080\text{ bytes}$ using binary convention. Both methods are shown below).
- **Total frame size:** 1600 bytes
- **Protocol overhead per frame:** 80 bytes

Calculations (Decimal Convention: $1\text{ MB} = 10^6\text{ bytes}$)

- **Useful data per frame:** $\text{Frame\ size}-\text{Overhead}=1600\text{\ bytes}-80\text{\ bytes}=1520\text{\ bytes}$
- **Total number of frames required:** $\frac{\text{Total\ useful\ data}}{\text{Useful\ data\ per\ frame}}=\frac{80,000,000\text{\ bytes}}{1520\text{\ bytes}}\approx 52,631.58\rightarrow \mathbf{52,632}\text{\ frames}$
- **Total overhead transmitted:** $\text{Total\ frames}\times \text{Overhead\ per\ frame}=52,632\times 80\text{\ bytes}=\mathbf{4,210,560}\text{\ bytes}$
- **Transmission efficiency:** $\frac{\text{Useful\ data\ per\ frame}}{\text{Total\ frame\ size}}\times 100=\frac{1520}{1600}\times 100=\mathbf{95\%}$

Calculations (Binary Convention: $1\text{ MB} = 2^{20}\text{ bytes}$)

- **Total number of frames required:** $\frac{83,886,080\text{\ bytes}}{1520\text{\ bytes}}\approx 55,188.21\rightarrow \mathbf{55,189}\text{\ frames}$
- **Total overhead transmitted:** $55,189\times 80\text{\ bytes}=\mathbf{4,415,120}\text{\ bytes}$
- **Transmission efficiency:** $\frac{1520}{1600}\times 100=\mathbf{95\%}$

Impact of Protocol Overhead on Network Performance

- **Reduces Throughput:** Overhead consumes bandwidth that could otherwise carry actual user data.
- **Increases Latency:** Processing extra header bytes at routers and switches adds to the propagation delay.
- **Wastes Bandwidth:** High overhead leads to low transmission efficiency, requiring more packets to send the same file.

Problem 10 Solution

Given Data

- **Original file size:** 240 MB
- **Compression rate:** 30% reduction

- **Transport Layer segment size:** 1400 bytes of payload data per segment
- **Data Link Layer overhead:** 60 bytes per frame

Calculations (Decimal Convention: $1\text{ MB} = 10^6\text{ bytes}$)

- **Compressed file size:** $240\text{\ MB}\times (1-0.30)=\mathbf{168}\text{\ MB}\quad (168,000,000\text{\ bytes})$
- **Total number of segments transmitted:** $\frac{168,000,000\text{\ bytes}}{1400\text{\ bytes}}=\mathbf{120,000}\text{\ segments}$
- **Total protocol overhead introduced:** $\text{Total\ frames/segments}\times \text{Overhead\ per\ frame}=120,000\times 60\text{\ bytes}=\mathbf{7,200,000}\text{\ bytes}$
- **Final amount of data transmitted across the Physical Layer:** $\text{Compressed\ data}+\text{Total\ overhead}=168,000,000+7,200,000=\mathbf{175,200,000}\text{\ bytes}\quad (175.2\text{\ MB})$

Calculations (Binary Convention: $1\text{ MB} = 2^{20}\text{ bytes}$)

- **Compressed file size:** $240\times 1,048,576\times 0.70=\mathbf{176,160,768}\text{\ bytes}\quad (168\text{\ MiB})$
- **Total number of segments transmitted:** $\frac{176,160,768\text{\ bytes}}{1400\text{\ bytes}}\approx 125,829.12\rightarrow \mathbf{125,830}\text{\ segments}$
- **Total protocol overhead introduced:** $125,830\times 60\text{\ bytes}=\mathbf{7,549,800}\text{\ bytes}$
- **Final amount of data transmitted across the Physical Layer:** $176,160,768+7,549,800=\mathbf{183,710,568}\text{\ bytes}$

How OSI Layers Work Together for Secure and Reliable Communication

- **Presentation Layer:** Compresses data for fast transit and encrypts it to protect sensitive medical records from unauthorized access.
- **Transport Layer:** Breaks massive files into manageable segments, assigning sequence numbers to guarantee reliable, in-order delivery and error recovery.
- **Data Link Layer:** Packages segments into network frames, handling local hardware addressing (MAC addresses) and physical error detection (FCS/CRC) to ensure data integrity over the immediate link.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

		and coherence			
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand